SEIT MASTER THEORY CANDIDATE

A Rigorous Partial Theoretical Framework, Honestly Titled Per Its Own Actual Closure Status

This document is NOT titled MASTER_THEORY_OF_EVERYTHING. No complete, verified Theory of Everything was derived by this campaign or by any prior phase of this project. This manuscript reports exactly what was actually derived, tested, falsified, and left open.

Abstract

This manuscript is the final research report of the Full SEIT Theory Derivation / Decounterfactualization / Compiler Execution Campaign. Its purpose was to determine, by actually executing code against this repository's own real infrastructure and real DESI DR1 data, whether four of the project's highest-leverage unresolved dependencies (H1: selection-principle well-posedness; H2: spectral-triple/Dirac-operator admissibility; H3: discrete-to-continuum geometric closure; H4: gauge-group derivation from G2/Spin(8) triality) could be genuinely closed. None achieved closure. H1-H3 remain OPEN or FAIL, backed by new negative evidence. H4's specific tested formulation is FALSIFIED by a decisive, general rank-counting theorem on compact Lie groups; a different, original formulation from this project's own historical corpus remains open but entirely unconstructed. The one branch of this project with a fully verified forward chain -- the graph-Laplacian spectral cascade -- is restated here as the strongest certified partial result. No fabricated closure, no reverse-fit constants, and no promotion of a hypothetical or externally-sourced claim to VERIFIED status appear anywhere in this document.

1. Introduction

This campaign followed two prior phases: (i) a real Master TOE Derivation Campaign that audited the project's full historical corpus and found no genuine unified theory, falsifying two specific claims (a fine-structure-constant non-sequitur and a reverse-fitted electron mass) in DTC COMPILER.docx; and (ii) an explicitly labeled counterfactual physics simulation constructing a hypothetical alternate-universe manuscript in which the project's open dependencies were assumed closed, paired with a real-vs-fiction gap matrix.

This campaign's task was to take the counterfactual manuscript's invented axioms seriously as hypotheses to actually test, using the real compiler, real code, and real data already present in this repository. Four hypotheses (H1-H4) were selected as the highest-leverage, most decidable open questions.

2. Methodology

- H1: static analysis of compiler/ir/forward_chain.py and compiler/backends/*.py for any implementation of Mathset, a persistence functional Pi(G), or a structural-cost functional S(G).
- H2: a real k-nearest-neighbour test graph (n=200, k=3), with L=D-W and D+=sqrt(L) computed via numpy.linalg.eigh, measuring sparsity and off-diagonal decay.
- H3: the real DESI DR1 LRG SGC pilot catalogue (160,150 points after redshift cut), reusing the project's existing sparse discrete-to-continuum pipeline at N=4000->8000, testing three candidate corrections to the known higher-mode non-convergence finding.
- H4: standard, cited dimension/rank facts for G2, Spin(8), SU(3), SU(2), U(1), combined with the maximal-torus theorem.

All four are registered in the canonical compiler with role=comparison. All 10 self-audits pass (0 issues each), and the full 95-test suite passes with no regressions.

3. Results

3.1 H1 -- Selection Closure: DOES NOT CLOSE

Mathset, Pi(G), and S(G) have zero implementation anywhere in this repository. G*=argmax_G Pi(G)/S(G) is not merely unproven -- its domain and objective function are undefined. This is a definitional gap, not a missing proof step.

3.2 H2 -- Spectral-Triple/Dirac Closure: DOES NOT CLOSE

D+=sqrt(L) is numerically dense (D+_sparsity_strict=1.0) even when L is sparse and local (L_sparsity=0.035, n=200, k=3). Off-diagonal decay is real but slow: 0.290, 0.280, 0.268 at graph-distances 1, 2, 3. A genuine Dirac-type operator requires [D,a] bounded for a local algebra element a; the operator square root does not preserve locality. KO-dimension of the natural real structure is 0 mod 8, not the required 6 mod 8.

3.3 H3 -- Discrete-to-Continuum/Geometry Closure: DOES NOT CLOSE

Against real DESI data (N=4000->8000): baseline shows 5/5 mode clusters unstable (subspace_cosine=0.9877). Tighter ARPACK tolerance reproduces the identical instability, ruling out solver precision. A 2x bandwidth increase stabilizes low modes [1,3] but leaves high modes [5,15] badly unstable (subspace_cosine=0.0195). The counterfactual manuscript's curvature-dependent kernel correction is ruled out analytically as circular: it requires the target Ricci scalar R(x) as a kernel input to derive R(x). FC005_CHECKPOINT.md's frozen result is extended, not overwritten.

3.4 H4 -- Gauge/Internal Algebra Closure: SPECIFIC CLAIM FALSIFIED

The claim that the triality-fixed subgroup of Spin(8) equals SU(3)xSU(2)xU(1) is mathematically impossible. The triality-fixed subgroup of Spin(8) is G2 (dim 14, rank 2). By the maximal-torus theorem, rank(SU(3)xSU(2)xU(1))=4 exceeds rank(G2)=2, so no embedding can exist. This is decisive and dimension-independent.

Distinct from the repository's own claim -- G=Aut(octonions)xSpin(8) superset SU(3)xSU(2)xU(1), a direct product -- where rank(G2xSpin(8))=6>=4, not ruled out by this argument, but with zero actual construction anywhere in this repository. It remains open, not falsified.

4. The Strongest Verified Partial Result

The graph-Laplacian spectral cascade, unchanged by this campaign, is the one branch of this project with a fully verified forward chain:

- G -> L = D - W (graph Laplacian, by construction)
- $L\ \phi_n = \lambda_n\ \phi_n$ (spectral theorem; VERIFIED against exact characteristic-polynomial eigenvalues for n<=8)
- $R(t) = e^{-tL}$ (heat semigroup; VERIFIED to <1e-6 residual)
- lim_(t->∞) R(t) = P_ker(L) (projector; VERIFIED to <1e-6 residual)

This chain is real, executed, tested, and representation-invariant. It is a mathematical fact about weighted graphs, not a physical theory, and no attempt is made to inflate its scope beyond that.

5. External Corpus Cross-Check

Two newly-supplied external workbooks were inspected as Layer C material. Their own OPEN-status sheets (e.g. '28_OPEN_PROPOSED_NO_GO') independently corroborate this campaign's H1 and H4 findings. Other sheets in the same corpus (e.g. '06_THEOREMS_CERTIFIED') assert sweeping 'CERTIFIED' status for a functorial continuum-limit theorem and a Standard-Model-emergence claim covering the same territory, and '07_PROOFS_COMPLETED' contains rows whose own 'Missing' column admits an incomplete proof while filed under a 'completed' heading. This is a second, independent instance of the claimed-closure-versus-actual-closure gap this campaign exists to prevent, and it is excluded from the canonical registries.

6. Discussion: Why No Theory of Everything Was Obtained

Three of the four gaps trace to one cause: this project has no working, non-circular selection principle. H1 shows the selection functional is undefined. H4's open residual (which specific embedding, and why) is itself a selection problem. H2 and H3 are independent technical obstructions -- operator locality and finite-sample convergence -- that would remain even if selection were solved.

7. Predictions

This campaign generates zero new falsifiable physical predictions. Full detail in MASTER_THEORY_PREDICTIONS.md.

8. Conclusion

No complete Theory of Everything was derived, and none is claimed. The strongest genuinely established result remains the graph-Laplacian spectral cascade. H1-H3 are OPEN/FAIL with new negative evidence; H4's specific tested formulation is FALSIFIED; its sibling claim remains open and unconstructed. This document is titled SEIT_MASTER_THEORY_CANDIDATE precisely because titling it otherwise would misrepresent what was actually derived.

Appendix A -- Machine-Readable Theory Object

Field	Value
status	NO COMPLETE TOE THEOREM OBTAINED
master_object	L (graph Laplacian) -- only fully VERIFIED forward chain
master_functional	none closes; spectral action strongest external candidate, blocked by H2
gauge_group	SU(3)xSU(2)xU(1) preserved as EXTERNAL established physics, not derived

Field	Value
geometry	OPEN -- Gate 2 never entered; DESI continuum limit FAIL/RETRIABLE
matter_content	OPEN -- no surviving particle-spectrum derivation

Appendix B -- Closure Status Registry Cross-Reference

MASTER_THEORY_CLOSURE_MATRIX.csv (23 rows), MASTER_THEORY_DEPENDENCY_DAG.json (105 nodes, 63 edges), MASTER_THEORY_EQUATION_REGISTRY.json (35 equations), MASTER_THEORY_PROOF_REGISTRY.json (7 proofs), MASTER_THEORY_FALSIFICATION_REGISTRY.json (7 records), MASTER_THEORY_PROVENANCE_REGISTRY.json (105 entries).